

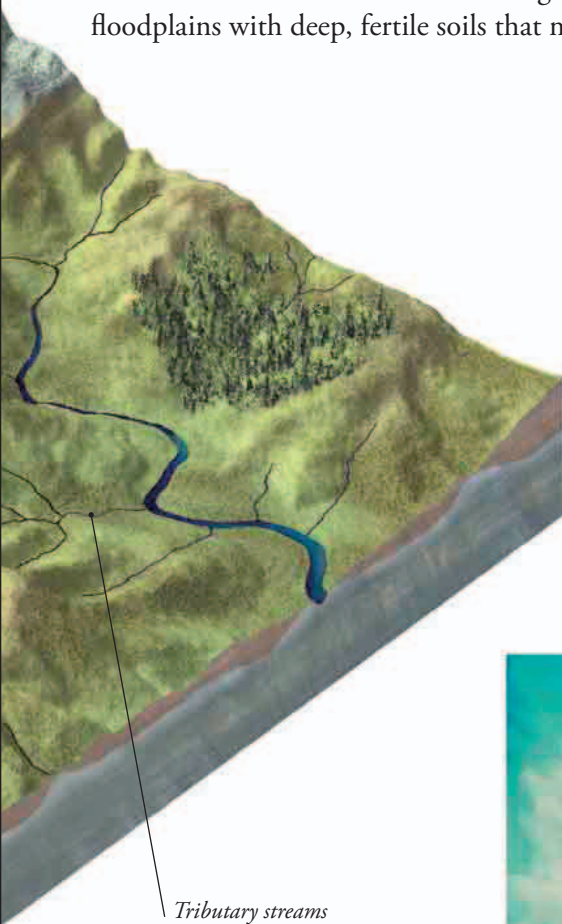
VALLEYS AND GORGES

Most river valleys in the uplands are winding V-shaped channels, often thick with trees. But some rivers flow through dramatic sheer-sided gorges. These can be created by the collapse of limestone cave systems, or by a river cutting down through a landscape that is slowly being pushed upward by the forces that build mountains.



FERTILE PLAINS

Rivers flowing from hills onto flatter ground can carry vast amounts of sediment. Heavy rain or spring snowmelt swells the rivers until they burst their banks and flood the surrounding landscape. Sediment settles from the slow-moving floodwaters to form broad floodplains with deep, fertile soils that make excellent farmland.



Tributary streams join the river and add to its flow



MEANDERS

A river tends to wind across its floodplain in loops called meanders. The flow cuts away the outside bank of the loop while depositing sediment on the inside bank, so the meanders become steadily more extreme. Eventually, the river may take a shortcut, leaving an isolated lake known as an oxbow.



▲ AMAZON OXBOW

Here, one of the meanders on the Amazon River has become an oxbow lake.

ESTUARIES AND DELTAS

When rivers reach the sea, the freshwater mixes with saltwater. This makes particles suspended in the water clump together and settle in thick layers of mud. These may form a muddy estuary, or fan out to sea in a delta. The river flow over a delta splits into many channels, as seen in this satellite view of the Yukon Delta in Alaska.

